

Multimodal AI for E-commerce Product Understanding

Name: _____

Day 1 Homework

My approach is to use the image to suggest a category and a short description, while keeping the seller in control when the model is unsure.

1. Input

The main input is one product image (JPG, PNG or WebP). At upload time I would check file type, size, corruption and minimum resolution. I would also check if the image is blurry, dark, heavily cropped, or does not contain a clear product.

Optional context such as the platform category list, seller title and extra images can improve the result, but the basic flow should still work with only one image.

2. Outputs

- Product category: a path from the approved taxonomy, for example Electronics > Audio > Headphones, plus a confidence score and top three choices. Low confidence should return Needs Review instead of forcing an answer.
- Product description: one or two factual sentences based on visible details. The model should not invent brand, material, price, battery life or other specifications that cannot be seen.

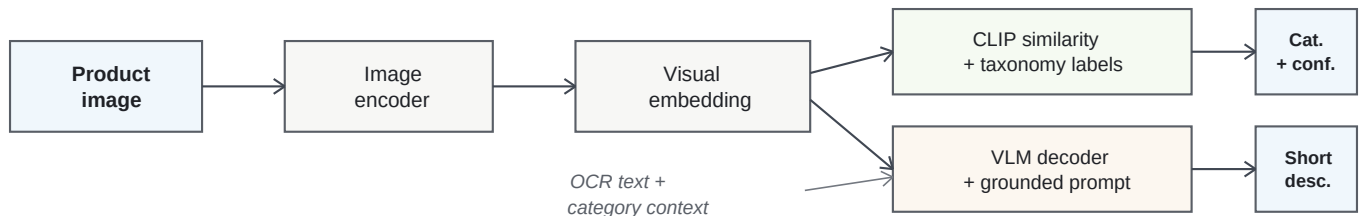
3. Model Selection

I would not depend on one free-form model for everything. For category prediction I would start with CLIP (or a fine-tuned vision encoder) and compare the image embedding with text embeddings of the allowed categories. For the description I would use BLIP-2 or a similar vision-language model. OCR can help with visible packaging text, but OCR text is only supporting evidence.

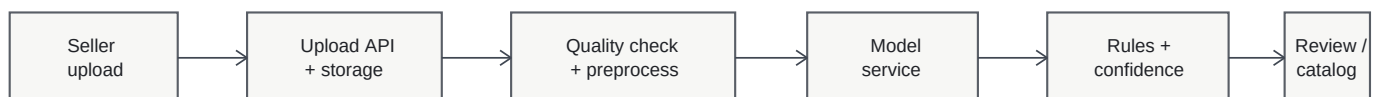
Why this split?

Category output stays controlled by the platform taxonomy, while the language model can write naturally. It also makes confidence thresholds, testing and replacement of one component easier.

Rough Model Architecture



Rough High-Level Design (HLD)



If confidence is low, the result goes to the seller or reviewer instead of being auto-published.

Main design decisions

Use a versioned category taxonomy; keep an Unknown/Other option; do not auto-publish uncertain results; and save seller corrections only after review so bad feedback does not directly train the model.

System Flow, Edge Cases and Evaluation

4. Step-by-Step Flow

1. Validate the upload and run image-quality and content-safety checks.
2. Detect the main product. If several products are equally important, ask the seller to crop or select one.
3. Preprocess the crop and run OCR when packaging or labels are visible.
4. Predict top-k categories, respecting the parent-child taxonomy, and calculate confidence.
5. Generate a short description using visual features, category context and strict instructions not to guess.
6. Run post-checks: unsupported claims, category-description conflict, length and restricted content.
7. Route by confidence: show a suggestion, offer top choices, or send to manual review.
8. Store the seller-approved result and correction for monitoring and later model improvement.

Practical edge cases and fallback

Bad/no product image Ask for a clearer image; do not guess.	OCR conflicts with image Treat text as weak evidence; flag conflict.
Multiple products Ask user to crop/select the main item.	Restricted item Send through policy checks/manual review.
Unknown/new product Return Other/Needs Review and log it.	Model timeout Retry safely, then allow manual entry.
Unclear brand/spec Omit it from the description.	Taxonomy/model drift Version labels and monitor by category.

5. Evaluation

- Offline test set: use human-checked images not used in training, with common and rare categories, different backgrounds, devices and image quality.
- Category: top-1/top-3 accuracy, macro F1, hierarchical error distance, unknown-product recall and confidence calibration. I would report results by category and difficult-image slice, not only one overall score.
- Description: human scoring for factuality, useful visible details, clarity and category consistency. Hallucination rate is a key metric. BLEU/ROUGE can be secondary because many descriptions can be correct.
- Production: p95 latency, failure and timeout rate, cost per image, review rate, seller acceptance/correction rate and data/model drift. I would start with a small pilot and compare it with manual listing.

6. Business Value

The system can reduce listing time and make categories and descriptions more consistent. Better metadata should improve search, filters, recommendations and inventory quality. I would measure listing time saved, fewer category corrections, seller completion rate, search relevance and conversion. Human confirmation protects customer trust.

Bonus - real-world example

Google Lens uses image and text understanding to identify products and show visually similar shopping results. It is useful when a customer can photograph an item but does not know the right words to search for it.